# Regulatory Complaint Investigation and CAPA Report

Confidential - AI Draft

| | |
|---|---|
| **Document No.** | DP-M4-20260708-DEVICESH |
| **Classification** | Confidential - AI Draft |
| **Generated** | 2026-07-08T09:48:14Z |
| **Device** | DocPlus+ Pulse Oximeter (DEV-PULSE-OX) |
| **Firmware** | v3.4 |
| **Framework** | ISO 13485 |
| **Prepared By** | DocPlus+ M4 Evidence and Documentation Layer |

This document was generated from live M1 graph facts and M2 agent outputs. Every summarized claim is tied to captured agent state, evidence citations, or audit findings.

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This output follows a formal regulatory-document pattern: controlled metadata first, then complaint facts, investigation logic, objective evidence, risk decision, CAPA plan, review gates, and approval placeholders.

# 1. Document Profile and Controlled Metadata

| | |
|---|---|
| **Document Title** | DocPlus+ Regulatory Complaint Investigation and CAPA Report |
| **Document Number** | DP-M4-20260708-DEVICESH |
| **Revision** | A - AI Generated Draft |
| **Complaint ID** | CPL-20260708-2FCB |
| **CAPA ID** | CAPA-20260708-42C9 |
| **RTM ID** | RTM-DP-2FCB42C9 |
| **Classification** | Confidential - AI Draft for quality and regulatory review |
| **Generated Timestamp** | 2026-07-08T09:48:14Z |
| **System of Record** | DocPlus+ M4 Evidence and Documentation Layer |
| **Device** | DocPlus+ Pulse Oximeter (DEV-PULSE-OX) |
| **Current Firmware** | v3.4 |
| **Affected Component** | COMP-MEASUREMENT |
| **Regulatory Framework** | ISO 13485 |

## Data Quality and Source Provenance

This report references 47 graph fact(s): 15 controlled, 24 extracted, 5 inferred, 3 synthetic.

Uncontrolled graph facts appear in this report and are visibly tagged. [DEMO DATA - NOT CONTROLLED EVIDENCE] means the item cannot support CAPA closure without controlled verification. [INFERRED DATA - REVIEW REQUIRED] means the item was inferred and requires reviewer confirmation before final use.

| Source Type | Referenced Fact Count |
|---|---|
| controlled | 15 |
| extracted | 24 |
| inferred | 5 |
| synthetic | 3 |
| internal | 0 |

## Document Purpose

This report documents the complaint intake, AI-assisted investigation record, retrieved objective evidence, risk decision, CAPA recommendations, audit-style gaps, and corrective action controls. It is intended as a controlled draft that a Quality or Regulatory reviewer can verify, approve, or return for correction.

# 2. Regulatory Format Controls

## ALCOA+ Data Integrity Mapping

| Principle | DocPlus+ Implementation |
|---|---|
| Attributable | Prepared by DocPlus+ M4; generated timestamp and device scope are recorded. |
| Legible | A4 PDF with selectable text, structured headings, repeatable tables, and page numbers. |
| Contemporaneous | Generated at the time of complaint-report request from live agent state. |
| Original | Marked as AI Draft until Quality and Regulatory reviewers approve. |
| Accurate | Risk, CAPA, evidence, audit, and trace sections are populated from structured state. |
| Complete | Includes intake, investigation, evidence, risk, CAPA, audit, trace, and approval fields. |
| Consistent | Uses the same DocPlus+ controlled layout for all complaint-report exports. |
| Enduring | Saved as a PDF artifact under the application output directory. |
| Available | Downloadable through the M4 document endpoint and traceable by document number. |

## Applicable Regulatory Basis

| Framework | Relevance to This Report |
|---|---|
| ISO 13485 | Complaint handling, CAPA discipline, documented information, traceable quality records. |
| 21 CFR Part 820 | Design controls, complaint files, CAPA, and objective evidence expectations. |
| EU MDR | Technical documentation, post-market surveillance, vigilance, and risk-benefit continuity. |
| 21 CFR Part 11 | Electronic record controls, auditability, attribution, and durable retrieval. |

# 3. Executive Summary

| | | |
|---|---|---|
| **Severity**<br>**High** | **Risk**<br>**Critical / RPN 20** | **Reportable**<br>**True** |
| **Evidence Confidence**<br>**high (score 1.00)** | **Evidence Class Mix**<br>**Controlled current verification: 12** | **Evidence Items**<br>**12** |
| **Audit Findings**<br>**1** | **Trace Decay Alerts**<br>**1** | |

**Complaint:** Device shows low readings.

**Agent conclusion:** Complaint: DEV-PULSE-OX reports low readings (severity: High). The affected component is COMP-MEASUREMENT. Among assessed hypotheses, calibration drift in the measurement component (HYP-001, base probability 0.43) is the leading candidate, supported by a closed, confirmed device-failure incident (CPL-100-035: calibration drift following lens carrier creep). Environmental (HYP-002, 0.32) and user-interference factors (HYP-003, 0.25) remain plausible given similar incidents (low temperature, nail polish, direct sunlight), but no direct evidence links this complaint to those conditions. Multiple controlled verification runs for firmware v3.4 (e.g., REQ/TC-MEAS-035, -038, -040, -023, -026) show Pass, reducing the likelihood of a systematic design nonconformance. With no firmware/serial/lot or objective evidence provided for this specific unit, the root cause cannot be confirmed; prioritized hypotheses are HYP-001 > HYP-002 > HYP-003 pending returned-unit and use-environment data.

## Risk Evidence Confidence Basis

Weighted evidence-class score 1.00 from 12 item(s): Controlled current verification=12. CAPA-blocking evidence-class item count: 0. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Confidence drivers: RUN-TC-MEAS-040-V3-4: controlled_verification, RUN-TC-MEAS-035-V3-4: controlled_verification, RUN-TC-MEAS-038-V3-4: controlled_verification, RUN-TC-MEAS-023-V3-4: controlled_verification, RUN-TC-MEAS-026-V3-4: controlled_verification, RUN-TC-MEAS-031-V3-4: controlled_verification, RUN-TC-MEAS-036-V3-4: controlled_verification, RUN-TC-MEAS-037-V3-4: contro...

## Quality Review Decision

The complaint should remain open until objective evidence is reconciled, CAPA ownership is assigned, and any major or critical audit findings affecting the complaint scope are remediated or justified.

# 4. Complaint Investigation Record

| Field | Generated Value | Regulatory Basis |
|---|---|---|
| Complaint ID | CPL-20260708-2FCB | ISO 13485 8.2.2 |
| Date Received | 2026-07-08T09:48:14Z | 21 CFR Part 11 |
| Device UDI | UDI-DI-08717648-532226 | EU MDR Article 27 / 21 CFR 830 |
| Model / Version | DOCPLUS+-PULSE-OXIMETER / v3.4 | IEC 62304 Section 8 |
| Complaint Source | Post-market complaint intake | ISO 13485 8.2.2 |
| Complaint Narrative | Device shows low readings. | 21 CFR 820 complaint file |
| Patient Impact | Potential incorrect or unreliable information available to the user. | EU MDR Article 87 |
| MDR Reportable? | Yes | EU MDR Article 87 / 21 CFR 803 |
| CAPA Initiated? | Yes - CAPA-20260708-42C9 | 21 CFR 820.100 |
| Affected Component | COMP-MEASUREMENT | ISO 13485 8.5.2 |
| Closure Status | Closed: The scoped evidence chain contains 12 current controlled verification record(s) and no simulated, synthetic, candidate, historical, or no-evidence items. | ISO 13485 8.2.2 |

## Reportability Decision Logic

| Condition | DocPlus+ Assessment | Action |
|---|---|---|
| Death or serious injury | Not stated in complaint text. | Monitor for escalation during intake and investigation. |
| Potential harm or near miss | Potential incorrect or unreliable information available to the user. | Treat as reportability review required. |
| Device relationship | Complaint maps to Measurement and Sensor Performance. | Keep CAPA open until engineering evidence confirms or rules out relationship. |
| Malfunction/use issue | Reported condition indicates possible device malfunction or use-related failure. | Verify recurrence risk and effectiveness of controls. |

## Related Prior Complaints

| Complaint ID | Similarity | Summary | Source |
|---|---|---|---|
| CMP-001 | 2 | Patient reported intermittent low SpO2 readings during home monitoring. [DEMO DATA - NOT CONTROLLED EVIDENCE] | synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE] |
| CMP-008 | 2 | Field report indicates inconsistent readings in low temperature environment. [DEMO DATA - NOT CONTROLLED EVIDENCE] | synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE] |
| CPL-100-035 | 2 | Readings differ significantly from clinical device. [INFERRED DATA - REVIEW REQUIRED] | inferred [INFERRED DATA - REVIEW REQUIRED] |
| CPL-100-038 | 2 | Nail polish causes inaccurate readings. [INFERRED DATA - REVIEW REQUIRED] | inferred [INFERRED DATA - REVIEW REQUIRED] |
| CPL-100-041 | 2 | Bright sunlight interferes with readings. [INFERRED DATA - REVIEW REQUIRED] | inferred [INFERRED DATA - REVIEW REQUIRED] |

# 5. Requirements Traceability Matrix

| Req ID | Requirement | Scope / Source | Relevance | Evidence | Test Case | Status / Result |
|---|---|---|---|---|---|---|
| REQ-MEAS-035 | Readings differ significantly from clinical device. | strong - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 8. | verified_current - Controlled current verification | TC-MEAS-035 | current and usable - Pass |
| REQ-MEAS-026 | Device continuously searches for signal. | strong - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 6. | verified_current - Controlled current verification | TC-MEAS-026 | current and usable - Pass |
| REQ-MEAS-029 | Device reports impossible SpO2 values. | strong - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 6. | verified_current - Controlled current verification | TC-MEAS-029 | current and usable - Pass |
| REQ-MEAS-032 | Perfusion index is missing. | strong - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 6. | stale - Historical controlled evidence - stale or prior version | TC-MEAS-032 | stale - Pass |
| REQ-MEAS-033 | Signal quality indicator always shows poor. | strong - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 6. | verified_current - Controlled current verification | TC-MEAS-033 | current and usable - Pass |
| REQ-MEAS-034 | Device cannot detect weak pulse. | strong - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 6. | verified_current - Controlled current verification | TC-MEAS-034 | current and usable - Pass |
| REQ-MEAS-038 | Nail polish causes inaccurate readings. | strong - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 6. | verified_current - Controlled current verification | TC-MEAS-038 | current and usable - Pass |
| REQ-MEAS-041 | Bright sunlight interferes with readings. | strong - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 6. | verified_current - Controlled current verification | TC-MEAS-041 | current and usable - Pass |
| REQ-MEAS-021 | SpO2 reading remains at 0%. | medium - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 4. | verified_current - Controlled current verification | TC-MEAS-021 | current and usable - Pass |
| REQ-MEAS-022 | Pulse rate is not detected. | medium - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 4. | verified_current - Controlled current verification | TC-MEAS-022 | current and usable - Pass |
| REQ-MEAS-023 | SpO2 reading fluctuates excessively. | medium - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 4. | verified_current - Controlled current verification | TC-MEAS-023 | current and usable - Pass |
| REQ-MEAS-024 | Pulse rate reading jumps randomly. | medium - RTM_Master.xlsx | Matched complaint terms against measurement_core; relevance score 4. | verified_current - Controlled current verification | TC-MEAS-024 | current and usable - Pass |

## Trace Decay Risk Statement

Rows marked STALE are not filler rows: they are generated from the live Trace Decay result and indicate verification links that require review before the device can claim current-firmware evidence.

# 6. Design History File Index

| # | Element | Document ID | Generated Unique Content | Status |
|---|---|---|---|---|
| 1 | Planning | DDP-DP-2FCB42C9 | Design plan for complaint-impact review | Open |
| 2 | Design Input | URS-DP-2FCB42C9 | Complaint-linked requirement review for Device shows low readings | Draft |
| 3 | Design Output | SDD-DP-2FCB42C9 | Measurement and Sensor Performance design output references mapped | Draft |
| 4 | Design Review | DRM-DP-2FCB42C9 | Cross-functional review required | Pending |
| 5 | Verification | VPR-DP-2FCB42C9 | Objective evidence required for Measurement and Sensor Performance on v3.4 | Open |
| 6 | Validation | VAL-DP-2FCB42C9 | Use-condition validation required for Device shows low readings | Open |
| 7 | Design Transfer | DTR-DP-2FCB42C9 | Release readiness blocked until evidence reconciled | Pending |
| 8 | Design Changes | DCR-DP-2FCB42C9 | Change/service impact assessment linked to Measurement and Sensor Performance | Open |
| 9 | DHF Index | DHFI-DP-2FCB42C9 | DocPlus+ index for DocPlus+ Pulse Oximeter | Generated |

## Validation Gap Check

The validation section remains open because the complaint describes a user-reported condition involving Measurement and Sensor Performance. DocPlus+ therefore does not mark the DHF complete until validation, verification, or justified non-reproducibility evidence is linked.

# 7. EU MDR Technical Documentation - Annex II / III

| Annex Ref | Section | DocPlus+ Generated Content |
|---|---|---|
| Annex II Section 1 | Device Description | UDI UDI-DI-08717648-532226 scoped to complaint involving Measurement and Sensor Performance. |
| Annex II Section 3 | Design and Manufacturing Information | Design/change impact for Measurement and Sensor Performance is carried into design output review. |
| Annex II Section 4 | GSPR Compliance | Reported condition maps to safety/performance review: Device shows low readings. |
| Annex II Section 5 | Benefit-Risk Analysis | Critical risk complaint requires residual-risk review before closure. |
| Annex II Section 6.1 | Verification and Validation | Objective verification evidence must be reconciled against the complaint scope. |
| Annex III Section 1.3 | PMS / PSUR Input | Complaint is retained as a post-market signal for trend monitoring. |
| Annex III Section 1.5 | Vigilance Reporting | Reportability remains open until human regulatory review confirms decision. |

## Team NB Structure Check

The package preserves a cohesive structure with section titles, metadata, and cross-references so that information is locatable by a reviewer rather than buried in a generic AI narrative.

## Record Boundary

The report is generated only from live application state: M1 graph context, M2 agent outputs, M4 evidence retrieval, AuditShadow findings, and Trace Decay alerts. It does not certify regulatory closure until the signature block is completed by authorized reviewers.

# 8. Complaint Investigation Workflow and Execution Trace

## 8.1 Structured Complaint Intake Detail

| | |
|---|---|
| **Device ID** | DEV-PULSE-OX |
| **Affected Component** | COMP-MEASUREMENT (COMP-MEASUREMENT) |
| **Severity** | High |
| **Timeline** | not specified |
| **Symptom Codes** | SYMPTOM_MISSING, SYMPTOM_DEVICE, SYMPTOM_SHOWS, SYMPTOM_LOW, SYMPTOM_READINGS |
| **Match Terms** | device, readings |
| **Similar Prior Complaints** | 5 |

### Complaint Narrative

Device shows low readings.

### Related Prior Complaints

| Complaint ID | Similarity | Summary | Source |
|---|---|---|---|
| CMP-001 | 2 | Patient reported intermittent low SpO2 readings during home monitoring. [DEMO DATA - NOT CONTROLLED EVIDENCE] | synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE] |
| CMP-008 | 2 | Field report indicates inconsistent readings in low temperature environment. [DEMO DATA - NOT CONTROLLED EVIDENCE] | synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE] |
| CPL-100-035 | 2 | Readings differ significantly from clinical device. [INFERRED DATA - REVIEW REQUIRED] | inferred [INFERRED DATA - REVIEW REQUIRED] |
| CPL-100-038 | 2 | Nail polish causes inaccurate readings. [INFERRED DATA - REVIEW REQUIRED] | inferred [INFERRED DATA - REVIEW REQUIRED] |
| CPL-100-041 | 2 | Bright sunlight interferes with readings. [INFERRED DATA - REVIEW REQUIRED] | inferred [INFERRED DATA - REVIEW REQUIRED] |

## 8.2 Investigation Chronology

| Time | Agent | Action | Status | Detail |
|---|---|---|---|---|
| 2026-07-08T09:43:39.637682Z | langgraph_supervisor | route to complaint_intake | info | Recorded |
| 2026-07-08T09:43:39.637682Z | complaint_intake | extract structured complaint | started | Recorded |
| 2026-07-08T09:47:29.263315Z | complaint_intake | extract structured complaint | completed | Elapsed 229625.4 ms |
| 2026-07-08T09:47:29.263315Z | langgraph_supervisor | handoff check: similar incidents preserved for root cause | info | Structured event metadata recorded |
| 2026-07-08T09:47:29.315549Z | langgraph_supervisor | route to root_cause | info | Recorded |
| 2026-07-08T09:47:29.315549Z | root_cause | generate complaint-specific failure hypotheses | started | Recorded |
| 2026-07-08T09:47:34.840837Z | root_cause | generate complaint-specific failure hypotheses | completed | Elapsed 5525.9 ms |
| 2026-07-08T09:47:34.942752Z | langgraph_supervisor | route to firmware_traceability_ripple_check | info | Recorded |
| 2026-07-08T09:47:34.942752Z | firmware_traceability_ripple_check | run firmware traceability ripple checks | started | Recorded |
| 2026-07-08T09:47:34.944038Z | firmware_traceability_ripple_check | hypothesis capability check | info | Structured event metadata recorded |
| 2026-07-08T09:47:34.945108Z | firmware_traceability_ripple_check | hypothesis capability check | info | Structured event metadata recorded |
| 2026-07-08T09:47:34.946348Z | firmware_traceability_ripple_check | hypothesis capability check | info | Structured event metadata recorded |
| 2026-07-08T09:47:34.946348Z | firmware_traceability_ripple_check | run firmware traceability ripple checks | completed | Elapsed 4.0 ms |
| 2026-07-08T09:47:35.069838Z | langgraph_supervisor | route to evidence | info | Recorded |
| 2026-07-08T09:47:35.069838Z | evidence | retrieve and validate evidence | started | Recorded |
| 2026-07-08T09:47:36.406071Z | evidence | component-scoped evidence retrieval | info | Structured event metadata recorded |
| 2026-07-08T09:47:36.406071Z | evidence | retrieve and validate evidence | completed | Elapsed 1336.3 ms |
| 2026-07-08T09:47:36.566617Z | langgraph_supervisor | route to risk | info | Recorded |

## Chronology Interpretation

The sequence confirms the complaint passed through intake, root-cause hypothesis generation, firmware traceability ripple checking, evidence retrieval, risk scoring, CAPA drafting, and final reasoning. This preserves a reviewer-readable process trail for the AI investigation rather than presenting the CAPA as an unsupported narrative.

## Trace AI - Pipeline Observability

Trace AI is operational engineering telemetry for this report generation. It records pipeline execution, LLM metadata, and retrieval/tool metadata only. It is not clinical evidence, not regulatory objective evidence, and not the same as Trace Decay or the Firmware Traceability Ripple Check.

| | |
|---|---|
| **Total wall-clock time** | 764017.29 ms |
| **Agent steps** | 10 |
| **LLM calls** | 3 |
| **Tool/retrieval calls** | 23 |
| **Tokens** | 22144 total (15557 prompt / 6587 completion) |
| **Longest agent** | audit_shadow (263837.0 ms) |

## Agent Step Trace

| Step ID | Agent | Action | Status | Elapsed | Sub-events |
|---|---|---|---|---|---|
| step-98fb5816d2 | complaint_intake | extract structured complaint | success | 229625.4 ms | LLM 0; tools 4 |
| step-f8aef9ef11 | audit_shadow | run adversarial audit | success | 263837.0 ms | LLM 0; tools 5 |
| step-8fb5fb1748 | cybersecurity | scan SBOM components against NVD | success | 224541.81 ms | LLM 0; tools 3 |
| step-9737d69b12 | root_cause | generate complaint-specific failure hypotheses | success | 5525.86 ms | LLM 1; tools 2 |
| step-b440cc06f0 | firmware_traceability_ripple_... | run firmware traceability ripple checks | success | 3.98 ms | LLM 0; tools 0 |
| step-7b63bafb5a | evidence | retrieve and validate evidence | success | 1336.3 ms | LLM 0; tools 3 |
| step-c5662cff3e | risk | calculate ISO 14971 risk | success | 353.21 ms | LLM 0; tools 2 |
| step-42d99bc552 | capa | draft CAPA with citations | success | 4884.99 ms | LLM 0; tools 4 |
| step-86d64f995e | openai_reasoning | final complaint reasoning | success | 22788.8 ms | LLM 1; tools 0 |
| step-9198b38635 | openai_reasoning | final audit reasoning | success | 11119.94 ms | LLM 1; tools 0 |

## LLM Call Metadata

| Call ID | Agent | Task | Model | Status | Tokens | Latency |
|---|---|---|---|---|---|---|
| llm-9d9febecfc | root_cause | root_cause_hypotheses | gpt-4o-mini | success | 5843 (5266/577) | 5232.61 ms |
| llm-63d6a0e75f | openai_reasoning | complaint_final_reasoning | gpt-5 | success | 8758 (4764/3994) | 22787.18 ms |
| llm-f9fa227308 | openai_reasoning | audit_final_reasoning | gpt-5 | success | 7543 (5527/2016) | 11113.99 ms |

## Tool and Retrieval Call Metadata

| Call ID | Agent | Tool | Query / Target | Results | Latency |
|---|---|---|---|---|---|
| tool-0934d44f45 | complaint_intake | graph_tools.resolve_device_id | device_id=DEV-PULSE-OX | | 1368.28 ms |
| tool-91f45c8718 | audit_shadow | graph_tools.resolve_device_id | device_id=DEV-PULSE-OX | | 1377.79 ms |
| tool-b245a8c6f4 | cybersecurity | graph_tools.resolve_device_id | device_id=DEV-PULSE-OX | | 1354.23 ms |
| tool-a6f50ef16b | complaint_intake | graph_tools.resolve_device_id | device_id=DEV-PULSE-OX | | 1040.39 ms |
| tool-b4a06f2bb5 | audit_shadow | graph_tools.resolve_device_id | device_id=DEV-PULSE-OX | | 1097.59 ms |
| tool-63a34a769e | cybersecurity | graph_tools.resolve_device_id | device_id=DEV-PULSE-OX | | 1186.6 ms |
| tool-fd7aa61380 | complaint_intake | graph_tools.get_device_context | device_id=DEV-PULSE-OX | 15 | 223255.62 ms |
| tool-6837f4f03c | cybersecurity | graph_tools.get_device_context | device_id=DEV-PULSE-OX | 15 | 223145.75 ms |
| tool-e024c14fa7 | audit_shadow | graph_tools.get_device_context | device_id=DEV-PULSE-OX | 15 | 224513.24 ms |
| tool-61497067fb | audit_shadow | graph_tools.resolve_device_id | device_id=DEV-PULSE-OX | | 72.88 ms |
| tool-83c8ceeadb | complaint_intake | graph_tools.find_similar_complaints | term=Device shows low readings. SYMPTOM_MISSING SYMPTOM_DEVICE SYMPTOM_... | 5 | 144.84 ms |
| tool-461d714796 | root_cause | graph_tools.get_graph_neighborhood | node_id=COMP-MEASUREMENT | 457 | 169.66 ms |
| tool-84bbfe517c | root_cause | graph_tools.get_graph_neighborhood | node_id=COMP-MEASUREMENT | 34 | 121.65 ms |
| tool-3d08c58853 | evidence | graph_tools.resolve_device_id | device_id=DEV-PULSE-OX | | 59.96 ms |
| tool-232e9a7de6 | evidence | graph_tools.get_component_scoped_ev... | device_id=DEV-PULSE-OX; component_id=COMP-MEASUREMENT; component_name=C... | 12 | 862.56 ms |
| tool-81d264c196 | evidence | vector_tools.retrieve_documents_debug | device_id=DEV-PULSE-OX; query=Device shows low readings. COMP-MEASUREME... | 0 | 472.25 ms |
| tool-a9669c03a3 | risk | graph_tools.resolve_device_id | device_id=DEV-PULSE-OX | | 122.59 ms |
| tool-9e74e8d2f2 | risk | graph_tools.open_capa_count | device_id=DEV-PULSE-OX | 2 | 352.7 ms |
| tool-4d5fcc8b98 | capa | graph_tools.resolve_device_id | device_id=DEV-PULSE-OX | | 94.51 ms |
| tool-67536eddfe | capa | graph_tools.get_audit_scope | device_id=DEV-PULSE-OX | 120 | 1183.66 ms |

3 additional tool/retrieval call(s) are retained in structured Trace AI data.

# 9. Root Cause Analysis

| ID | Hypothesis | Probability | Affected Area | Key Why Chain |
|---|---|---|---|---|
| HYP-001 | COMP-MEASUREMENT: Calibration Drift in Measurement Component [INFERRED DATA - REVIEW REQUIRED] | 43% | COMP-MEASUREMENT | Low readings observed from device \| Calibration issues can lead to inaccurate measurements [INFERRED DATA - REVIEW REQUIRED] |
| HYP-002 | COMP-MEASUREMENT: Environmental Effects on Measurement Accuracy [DEMO DATA - NOT CONTROLLED EVIDENCE] | 32% | COMP-MEASUREMENT | Inconsistent readings noted during specific conditions \| Measurement accuracy can be influenced by ambient conditions [DEMO DATA - NOT CONTROLLED EVIDENCE] |
| HYP-003 | COMP-MEASUREMENT: Interference from User Factors [INFERRED DATA - REVIEW REQUIRED] | 25% | COMP-MEASUREMENT | Low readings may be due to interference from external factors on measurement sensors [INFERRED DATA - REVIEW REQUIRED] |

## HYP-001 Evidence Balance

| Evidence For | Calibrations may lag behind subsequent changes in measurement functionality due to updates or environmental factors. [INFERRED DATA - REVIEW REQUIRED] |
|---|---|
| Evidence Against / Gaps | No immediate historical evidence indicating previous calibration failures. [INFERRED DATA - REVIEW REQUIRED] |
| Similar Incident Cross-Check | CPL-100-035: Readings differ significantly from clinical device - indicates calibration issues [INFERRED DATA - REVIEW REQUIRED] |
| Probability Rationale | Evidence suggests prior calibration issues may relate to the same component type, requiring enhanced scrutiny. [INFERRED DATA - REVIEW REQUIRED] |

## HYP-002 Evidence Balance

| Evidence For | CMP-008 discusses inconsistent readings in low temperature environments. [DEMO DATA - NOT CONTROLLED EVIDENCE] |
|---|---|
| Evidence Against / Gaps | No direct evidence linking current device performance to specific environmental conditions. [DEMO DATA - NOT CONTROLLED EVIDENCE] |
| Similar Incident Cross-Check | CMP-008: Field report indicates inconsistent readings in low temperature environment - shows potential for environmental influence [DEMO DATA - NOT CONTROLLED EVIDENCE] |
| Probability Rationale | Environmental conditions have been known to affect device accuracy, aligning with current complaint but lacking direct correlation. [DEMO DATA - NOT CONTROLLED EVIDENCE] |

## HYP-003 Evidence Balance

| Evidence For | CPL-100-038: Nail polish causes inaccurate readings demonstrates user influence on performance. [INFERRED DATA - REVIEW REQUIRED] |
|---|---|
| Evidence Against / Gaps | No specific evidence from the current user's context about interference due to nail polish or other factors. [INFERRED DATA - REVIEW REQUIRED] |
| Similar Incident Cross-Check | CPL-100-038: Closed case regarding nail polish interference - highlights user influence on device readings [INFERRED DATA - REVIEW REQUIRED] |
| Probability Rationale | User-related factors have previously been documented to affect performance, which aligns with the symptom but needs further scrutiny. [INFERRED DATA - REVIEW REQUIRED] |

## Root Cause Rationale

**1.** The complaint states the device shows low readings; COMP-MEASUREMENT is identified as the affected component.

**2.** HYP-001 (Calibration drift) has the highest base probability (0.43) and is supported by similar incident CPL-100-035, closed as a confirmed device failure with root cause "Optical calibration drifted outside tolerance following lens carrier creep."

**3.** Design/verification evidence for firmware v3.4 shows Pass for REQ/TC-MEAS-035 (Readings differ significantly from clinical device) and other measurement robustness tests (REQ/TC-MEAS-038 nail polish, -040 dark skin tone, -023 excessive fluctuation, -026 signal search), suggesting no current evidence of a systemic design issue in v3.4.

**4.** HYP-002 (Environmental effects) is plausible (base probability 0.32) and aligns with CMP-008 (inconsistent readings in low temperature), but there is no direct evidence that this complaint occurred under such conditions.

**5.** HYP-003 (User interference) is plausible (base probability 0.25) and is consistent with CPL-100-038 (nail polish) and CPL-100-041 (bright sunlight), but no complaint details indicate these factors.

**6.** Given the absence of unit identifiers, firmware, lot, timeline, or objective evidence, the most supportable disposition is prioritized hypotheses with HYP-001 leading, pending returned-unit evaluation and use-environment clarification.

## Primary Root Cause Position

For this generated draft, the leading position is that the complaint is associated with Measurement and Sensor Performance and the evidence/control set tied to the reported condition. The position remains provisional until engineering review confirms or rules out the suspected cause with objective evidence.

# 10. Evidence and Provenance

| Source | Evidence Class | Support | Conf. | Excerpt | Citation |
|---|---|---|---|---|---|
| REQ-MEAS-040 / TC-MEAS-040 / RUN-TC-MEAS-040-V3-4 | Controlled current verification | Yes | 96% | REQ-MEAS-040: Dark skin tone affects accuracy. Test TC-MEAS-040: Verification test TC-MEAS-040; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-040-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4... | [Source: REQ-MEAS-040/TC-MEAS-040, Confidence: 0.96] |
| REQ-MEAS-035 / TC-MEAS-035 / RUN-TC-MEAS-035-V3-4 | Controlled current verification | Yes | 96% | REQ-MEAS-035: Readings differ significantly from clinical device. Test TC-MEAS-035: Verification test TC-MEAS-035; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-035-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100... | [Source: REQ-MEAS-035/TC-MEAS-035, Confidence: 0.96] |
| REQ-MEAS-038 / TC-MEAS-038 / RUN-TC-MEAS-038-V3-4 | Controlled current verification | Yes | 96% | REQ-MEAS-038: Nail polish causes inaccurate readings. Test TC-MEAS-038: Verification test TC-MEAS-038; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-038-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression F... | [Source: REQ-MEAS-038/TC-MEAS-038, Confidence: 0.96] |
| REQ-MEAS-023 / TC-MEAS-023 / RUN-TC-MEAS-023-V3-4 | Controlled current verification | Yes | 96% | REQ-MEAS-023: SpO2 reading fluctuates excessively. Test TC-MEAS-023: Verification test TC-MEAS-023; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-023-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v... | [Source: REQ-MEAS-023/TC-MEAS-023, Confidence: 0.96] |
| REQ-MEAS-026 / TC-MEAS-026 / RUN-TC-MEAS-026-V3-4 | Controlled current verification | Yes | 96% | REQ-MEAS-026: Device continuously searches for signal. Test TC-MEAS-026: Verification test TC-MEAS-026; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-026-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression... | [Source: REQ-MEAS-026/TC-MEAS-026, Confidence: 0.96] |

| Source | Evidence Class | Support | Conf. | Excerpt | Citation |
|---|---|---|---|---|---|
| REQ-MEAS-031 / TC-MEAS-031 / RUN-TC-MEAS-031-V3-4 | Controlled current verification | Yes | 96% | REQ-MEAS-031: Pulse strength indicator is inaccurate. Test TC-MEAS-031: Verification test TC-MEAS-031; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-031-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression F... | [Source: REQ-MEAS-031/TC-MEAS-031, Confidence: 0.96] |

## 10.1 Firmware Traceability Ripple Check

This section reports a firmware-traceability check: for each root-cause hypothesis, it examines whether linked verification test records match the complaint or candidate firmware version, or are stale/mismatched. This is a graph-based traceability signal, not a behavioral model or physical test of the device. It identifies where verification evidence may be out of date, not what the device would actually do under the reported failure condition.

| Hypothesis | Analysis Type | Result | Interpretation | Evidence Class |
|---|---|---|---|---|
| HYP-001 | Not applicable - measurement_calibration_or_physiology | not_run | HYP-001 evaluates COMP-MEASUREMENT: Calibration Drift in Measurement Component. The firmware traceability ripple check was not run because this failure mode maps to 'measurement_calibration_or_physiology', outside firmware-to-test traceability freshness rev... | Firmware traceability signal - informational, not controlled verification evidence |
| HYP-002 | Not applicable - environmental_or_physical_conditions | not_run | HYP-002 evaluates COMP-MEASUREMENT: Environmental Effects on Measurement Accuracy. The firmware traceability ripple check was not run because this failure mode maps to 'environmental_or_physical_conditions', outside firmware-to-test traceability freshness r... | Firmware traceability signal - informational, not controlled verification evidence |
| HYP-003 | Not applicable - measurement_calibration_or_physiology | not_run | HYP-003 evaluates COMP-MEASUREMENT: Interference from User Factors. The firmware traceability ripple check was not run because this failure mode maps to 'measurement_calibration_or_physiology', outside firmware-to-test traceability freshness review. This hy... | Firmware traceability signal - informational, not controlled verification evidence |

# 11. CAPA Report

| | |
|---|---|
| **Product** | DocPlus+ Pulse Oximeter |
| **Software Version** | v3.4 |
| **CAPA Status** | Closed |
| **Priority** | High |
| **Risk Classification** | Major - potential impact to Measurement and Sensor Performance |
| **Evidence Confidence** | high (score 1.00) |
| **Evidence Class Mix** | Controlled current verification: 12 |

## Risk Evidence Confidence Basis

Weighted evidence-class score 1.00 from 12 item(s): Controlled current verification=12. CAPA-blocking evidence-class item count: 0. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

## Controlled Closure Gate

The scoped evidence chain contains 12 current controlled verification record(s) and no simulated, synthetic, candidate, historical, or no-evidence items.

### Problem Statement

A high priority complaint was opened for DocPlus+ Pulse Oximeter. The user reported: Device shows low readings. In practical terms, the concern is that the device may provide information that is inaccurate, unstable, or difficult to rely on. The complaint record confirms the reported condition; the exact technical cause and final correction remain under investigation until objective evidence is reviewed.

Evidence class gate: Controlled current verification

### Complaint Summary

Reported condition: Device shows low readings. Affected area: Measurement and Sensor Performance. Reported timing or trigger: Not fully specified by complaint text. Observed symptom signals: inconsistent or inaccurate measurement readings. No patient or user injury is stated in the complaint text; the risk review must still confirm potential safety impact.

Evidence class gate: Controlled current verification

### Immediate Containment Actions

Preserve the complaint evidence for Measurement and Sensor Performance, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact.

Evidence class gate: Controlled current verification

**Investigation**

Relevant device requirements, verification records, risk controls, and open quality actions were reviewed for the complaint scope. The investigation focused on the reported condition and the affected area: Measurement and Sensor Performance. For this complaint the scoped area is Measurement and Sensor Performance, with emphasis on measurement accuracy, sensor signal quality, calibration, and verification evidence. Engineering and Quality reviewed the available design expectations, test records, risk controls, event evidence, and related quality actions. 1 verification record needs Quality review. The scoped verification evidence is available for Quality review. Risk evidence basis: Weighted evidence-class score 1.00 from 12 item(s): Controlled current verification=12. CAPA-blocking evidence-class item count: 0.

Evidence class gate: Controlled current verification

**Root Cause Analysis**

The current root-cause hypothesis is: The measurement component's calibration may have drifted outside of tolerances, leading to low readings. This is a probable cause, not a final confirmed cause, because the CAPA record still needs returned-unit or accessory review, measurement/signal testing, use-condition review, and approved verification evidence showing whether the suspected cause is truly responsible.

Evidence class gate: Controlled current verification

**Why CAPA Remains Open**

The CAPA closure gate is closed because all scoped evidence items are current controlled verification records. Quality and Regulatory reviewers should preserve the signed closure package and continue normal complaint monitoring. Risk evidence confidence is high (score 1.00); Low uncertainty: all 12 evidence item(s) are current controlled verification evidence. Closure gate result: closed. The scoped evidence chain is entirely current controlled verification evidence, with no simulated, synthetic, candidate, historical, or no-evidence items.

Evidence class gate: Controlled current verification

**Risk Assessment**

The main hazard is that the device may provide information that is inaccurate, unstable, or difficult to rely on. The current risk level is Critical and the complaint should remain in active risk review under ISO 13485 until patient impact, recurrence potential, and residual risk acceptability are documented. Risk evidence confidence is high (score 1.00); Low uncertainty: all 12 evidence item(s) are current controlled verification evidence.

Evidence class gate: Controlled current verification

**Corrective Actions**

Reproduce or simulate the reported condition for Measurement and Sensor Performance under the reported use conditions. Engineering and Quality should review measurement accuracy, sensor signal quality, calibration, and verification evidence, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Maintain the controlled evidence package and closure approvals in the quality record.

Evidence class gate: Controlled current verification

**Preventive Actions**

Update the design, service, or release checklist so future changes affecting Measurement and Sensor Performance include complaint-specific verification for inconsistent or inaccurate measurement readings. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition.

Evidence class gate: Controlled current verification

**Verification of Effectiveness**

Effectiveness is supported by current controlled verification of measurement accuracy and signal reliability; Quality should confirm the affected area (Measurement and Sensor Performance) remains compliant through complaint monitoring and approved record retention.

Evidence class gate: Controlled current verification

### CAPA Effectiveness Criteria

The CAPA is effective when controlled evidence shows measurement accuracy and signal reliability meets the approved specification, the closure package is signed, and no related complaint recurs during six months after the corrected release or field correction.

Evidence class gate: Controlled current verification

### Lessons Learned

Future complaint handling for Measurement and Sensor Performance should connect user-reported symptoms, objective evidence, risk controls, and verification records early. Similar changes or service decisions should include test coverage for the reported condition and review of whether existing controls remain effective.

Evidence class gate: Controlled current verification

### CAPA Closure

Closure decision: closed. The scoped evidence chain contains 12 current controlled verification record(s) and no simulated, synthetic, candidate, historical, or no-evidence items. The CAPA record should retain the controlled evidence chain, reviewer signatures, and complaint monitoring plan.

Evidence class gate: Controlled current verification

Technical evidence identifiers, requirement IDs, audit findings, and source citations are retained in the evidence and audit appendices. They are not repeated here so the CAPA remains readable for Quality, Engineering, Regulatory, and non-technical reviewers.

## 11.1 CAPA Action Plan

| | |
|---|---|
| **CAPA Record** | CAPA-20260708-42C9 |
| **Complaint Reference** | CPL-20260708-2FCB |
| **CAPA Type** | Corrective and preventive action |
| **Current Status** | Closed |
| **Controlled Closure Gate** | The scoped evidence chain contains 12 current controlled verification record(s) and no simulated, synthetic, candidate, historical, or no-evidence items. |

## Action Plan

| Purpose | Owner | Required Action | How Closure Is Proven |
|---|---|---|---|
| Immediate measures | Quality / Support | Preserve the complaint evidence for Measurement and Sensor Performance, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact. | Evidence is preserved and the controlled closure package remains traceable. |
| Fix the issue | Engineering | Reproduce or simulate the reported condition for Measurement and Sensor Performance under the reported use conditions. Engineering and Quality should review measurement accuracy, sensor signal quality, calibration, and verification evidence, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Maintain the controlled evidence package and closure approvals in the quality record. | The confirmed cause is fixed or ruled out with current controlled verification evidence. |
| Prevent recurrence | Engineering / Quality | Update the design, service, or release checklist so future changes affecting Measurement and Sensor Performance include complaint-specific verification for inconsistent or inaccurate measurement readings. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition. | Future changes or service decisions retain the approved controls before release. |
| Check effectiveness | Quality Assurance | Effectiveness is supported by current controlled verification of measurement accuracy and signal reliability; Quality should confirm the affected area (Measurement and Sensor Performance) remains compliant through complaint monitoring and approved record retention. | Current controlled verification and monitoring support closure. |

## Acceptance Criteria for Closure

| Closure Gate | Required Evidence |
|---|---|
| Reported condition | The reported condition is corrected or ruled out in the controlled closure package. |
| Affected function | Measurement accuracy and signal reliability is verified by current controlled evidence. |
| Verification | All scoped evidence items are controlled current verification records. |
| Complaint monitoring | No related recurrence is observed during the defined follow-up period. |
| Approval | Quality and Regulatory reviewers maintain the approved completed CAPA record. |

# 12. Cybersecurity / SBOM Narrative

This section is device-level cybersecurity surveillance. It is not complaint evidence and it does not change the Risk Agent score automatically. It records what the Cybersecurity Agent found by reading the SBOM and querying the National Vulnerability Database for the CPE candidates in the workbook.

| | |
|---|---|
| **SBOM components** | 35 |
| **Components queried** | 25 |
| **NVD findings** | 7 |
| **Query date** | 2026-07-04 |
| **Source type** | extracted |
| **Cache status** | cache_hit |

## Executive Severity Rollup

| Critical | High | Medium | Low | None |
|---|---|---|---|---|
| 0 | 2 | 5 | 0 | 0 |

VEX-style exploitability status is shown per CVE. The current automated SBOM/NVD match does not include device-specific exploitability determinations, so findings default to Under Investigation until Engineering documents whether the vulnerable code path is affected, fixed, not affected, or not reachable in this device configuration.

## SBOM Component Coverage

| Component | Version | Supplier | CPE Candidate | NVD Result |
|---|---|---|---|---|
| Bootloader | 1.8.0 | Device OEM / Secure Boot Team | cpe:2.3:o:device_oem:pulseox_bootloader:1.8.0:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Embedded Firmware | 2.4.3 | Device OEM / Embedded Software Team | cpe:2.3:o:device_oem:pulseox_firmware:2.4.3:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Operating System / RTOS | 10.5.1 | Amazon Web Services, Inc. | cpe:2.3:o:amazon:freertos:10.5.1:*:*:*:*:*:*:* | open_cves_found: 1 CVE record(s) returned by NVD for this CPE. |
| Device Drivers | 1.6.2 | Device OEM / Driver Integration Team | cpe:2.3:a:device_oem:pulseox_device_drivers:1.6.2:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Signal Processing Software | 3.2.1 | Device OEM / Clinical Algorithm Supplier | cpe:2.3:a:device_oem:pulseox_signal_processing:3.2.1:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Measurement Control Logic | 2.1.0 | Device OEM / Embedded Software Team | cpe:2.3:a:device_oem:pulseox_measurement_control:2.1.0:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Alarm Management Module | 2.0.5 | Device OEM / Safety Software Team | cpe:2.3:a:device_oem:pulseox_alarm_manager:2.0.5:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| User Interface Module | 1.7.4 | Device OEM / UI Software Team | cpe:2.3:a:device_oem:pulseox_ui_module:1.7.4:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Settings Configuration Manager | 1.9.1 | Device OEM / Embedded Software Team | cpe:2.3:a:device_oem:pulseox_settings_manager:1.9.1:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Data Storage Manager | 1.5.0 | Device OEM / Data Platform Team | cpe:2.3:a:device_oem:pulseox_data_storage_manager:1.5.0:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Patient Record Manager | 1.3.2 | Device OEM / Data Platform Team | cpe:2.3:a:device_oem:pulseox_patient_record_manager:1.3.2:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |

| Component | Version | Supplier | CPE Candidate | NVD Result |
|---|---|---|---|---|
| Communication Stack | 2.2.0 | Device OEM / Connectivity Team | cpe:2.3:a:device_oem:pulseox_communication_stack:2.2.0:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Bluetooth Stack | 5.3.0 | Nordic Semiconductor ASA | cpe:2.3:a:nordicsemi:bluetooth_low_energy_stack:5.3.0:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Wi-Fi Stack | 2.10 | w1.fi project / Device OEM Integration | cpe:2.3:a:w1.fi:wpa_supplicant:2.10:*:*:*:*:*:*:* | open_cves_found: 1 CVE record(s) returned by NVD for this CPE. |
| TCP/IP Stack | 2.1.3 | lwIP project | cpe:2.3:a:lwip_project:lwip:2.1.3:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| USB Interface Software | 1.4.1 | Device OEM / Connectivity Team | cpe:2.3:a:device_oem:pulseox_usb_interface:1.4.1:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| File System | 3.8.1 | littlefs project / Device OEM Integration | cpe:2.3:a:littlefs_project:littlefs:3.8.1:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Logging Module | 1.2.8 | Device OEM / Platform Software Team | cpe:2.3:a:device_oem:pulseox_logging_module:1.2.8:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Error Handling Module | 1.1.6 | Device OEM / Platform Software Team | cpe:2.3:a:device_oem:pulseox_error_handling:1.1.6:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Time Management Module | 1.0.9 | Device OEM / Platform Software Team | cpe:2.3:a:device_oem:pulseox_time_manager:1.0.9:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Self-Test Module | 1.3.5 | Device OEM / Test Software Team | cpe:2.3:a:device_oem:pulseox_self_test:1.3.5:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Firmware Update Manager | 2.0.2 | Device OEM / Release Engineering Team | cpe:2.3:a:device_oem:pulseox_update_manager:2.0.2:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Authentication Module | 1.2.0 | Device OEM / Security Software Team | cpe:2.3:a:device_oem:pulseox_authentication_module:1.2.0:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |

| Component | Version | Supplier | CPE Candidate | NVD Result |
|---|---|---|---|---|
| Authorization Module | 1.2.0 | Device OEM / Security Software Team | cpe:2.3:a:device_oem:pulseox_authorization_module:1.2.0:*:*:*:*:*:*:* | no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04. |
| Cryptographic Library | 3.0.13 | OpenSSL Software Foundation | cpe:2.3:a:openssl:openssl:3.0.13:*:*:*:*:*:*:* | open_cves_found: 5 CVE record(s) returned by NVD for this CPE. |

## NVD CVE Findings - Structured Vulnerability Records

## Vulnerability Record - CVE-2024-28115

| | |
|---|---|
| **CVE ID / NVD Reference** | **CVE-2024-28115**<br>https://nvd.nist.gov/vuln/detail/CVE-2024-28115 |
| **CVSS Score / Severity / Vector** | HIGH / CVSS 8.8 / v3.1<br>Vector: CVSS:3.1/ AV:L/ AC:L/ PR:L/ UI:N/ S:C/ C:H/ I:H/ A:H |
| **CWE ID(s)** | CWE-284: Improper Access Control, NVD-CWE-Other: Other weakness classification |
| **Affected Component + Version** | Operating System / RTOS 10.5.1 |
| **Vulnerability Description** | NVD classifies the CVE under CWE-284, NVD-CWE-Other; review the public description for deployment-specific impact. |
| **Exploitability / Exposure Context** | Local or physically mediated exploitation should be assessed against service, update, or peripheral workflows. SBOM exposure: No direct exposure unless network stacks or external interfaces run as RTOS tasks. |
| **Recommended Controls** | review vendor fix status, confirm exploit preconditions, and document whether compensating controls cover the affected interface. SBOM controls: Minimal RTOS configuration, stack overflow checks, watchdog, task priority review, patch monitoring. |
| **VEX-Style Status** | Under Investigation |
| **Status** | Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM. |
| **Source / Reference** | extracted; NVD CVE API 2.0<br>https://nvd.nist.gov/vuln/detail/CVE-2024-28115 |

## Vulnerability Record - CVE-2023-52160

| | |
|---|---|
| **CVE ID / NVD Reference** | **CVE-2023-52160**<br>https://nvd.nist.gov/vuln/detail/CVE-2023-52160 |
| **CVSS Score / Severity / Vector** | MEDIUM / CVSS 6.5 / v3.1<br>Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:R/ S:U/ C:H/ I:N/ A:N |
| **CWE ID(s)** | CWE-287: Improper Authentication |
| **Affected Component + Version** | Wi-Fi Stack 2.10 |
| **Vulnerability Description** | NVD describes an authentication or access-control bypass condition in the affected component. |
| **Exploitability / Exposure Context** | Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Wi-Fi radio, WPA2/WPA3 negotiation, outbound cloud network. |
| **Recommended Controls** | update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: WPA2/WPA3, credential encryption, no ad-hoc mode, network allow-list, update monitoring. |
| **VEX-Style Status** | Under Investigation |
| **Status** | Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM. |
| **Source / Reference** | extracted; NVD CVE API 2.0<br>https://nvd.nist.gov/vuln/detail/CVE-2023-52160 |

## Vulnerability Record - CVE-2009-1390

| | |
|---|---|
| **CVE ID / NVD Reference** | **CVE-2009-1390**<br>https://nvd.nist.gov/vuln/detail/CVE-2009-1390 |
| **CVSS Score / Severity / Vector** | MEDIUM / CVSS 6.8 / v2.0<br>Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P |
| **CWE ID(s)** | CWE-287: Improper Authentication |
| **Affected Component + Version** | Cryptographic Library 3.0.13 |
| **Vulnerability Description** | NVD describes incomplete TLS certificate-chain validation, where a connection may be accepted because one certificate in an otherwise untrusted chain is trusted. |
| **Exploitability / Exposure Context** | Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features. |
| **Recommended Controls** | verify full certificate-chain validation, require trusted root and hostname match, and regression-test rejected-chain cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review. |
| **VEX-Style Status** | Under Investigation |
| **Status** | Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM. |
| **Source / Reference** | extracted; NVD CVE API 2.0<br>https://nvd.nist.gov/vuln/detail/CVE-2009-1390 |

## Vulnerability Record - CVE-2009-3765

| | |
|---|---|
| **CVE ID / NVD Reference** | **CVE-2009-3765**<br>https://nvd.nist.gov/vuln/detail/CVE-2009-3765 |
| **CVSS Score / Severity / Vector** | MEDIUM / CVSS 6.8 / v2.0<br>Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P |
| **CWE ID(s)** | CWE-310: Cryptographic Issues |
| **Affected Component + Version** | Cryptographic Library 3.0.13 |
| **Vulnerability Description** | NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch. |
| **Exploitability / Exposure Context** | Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features. |
| **Recommended Controls** | reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review. |
| **VEX-Style Status** | Under Investigation |
| **Status** | Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM. |
| **Source / Reference** | extracted; NVD CVE API 2.0<br>https://nvd.nist.gov/vuln/detail/CVE-2009-3765 |

## Vulnerability Record - CVE-2009-3766

| | |
|---|---|
| **CVE ID / NVD Reference** | **CVE-2009-3766**<br>https://nvd.nist.gov/vuln/detail/CVE-2009-3766 |
| **CVSS Score / Severity / Vector** | MEDIUM / CVSS 6.8 / v2.0<br>Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P |
| **CWE ID(s)** | CWE-310: Cryptographic Issues |
| **Affected Component + Version** | Cryptographic Library 3.0.13 |
| **Vulnerability Description** | NVD describes a certificate or TLS validation weakness in the affected secure-communication path. |
| **Exploitability / Exposure Context** | Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features. |
| **Recommended Controls** | update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review. |
| **VEX-Style Status** | Under Investigation |
| **Status** | Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM. |
| **Source / Reference** | extracted; NVD CVE API 2.0<br>https://nvd.nist.gov/vuln/detail/CVE-2009-3766 |

## Vulnerability Record - CVE-2009-3767

| | |
|---|---|
| **CVE ID / NVD Reference** | **CVE-2009-3767**<br>https://nvd.nist.gov/vuln/detail/CVE-2009-3767 |
| **CVSS Score / Severity / Vector** | MEDIUM / CVSS 4.3 / v2.0<br>Vector: AV:N/ AC:M/ Au:N/ C:N/ I:P/ A:N |
| **CWE ID(s)** | CWE-295: Improper Certificate Validation |
| **Affected Component + Version** | Cryptographic Library 3.0.13 |
| **Vulnerability Description** | NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch. |
| **Exploitability / Exposure Context** | Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features. |
| **Recommended Controls** | reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review. |
| **VEX-Style Status** | Under Investigation |
| **Status** | Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM. |
| **Source / Reference** | extracted; NVD CVE API 2.0<br>https://nvd.nist.gov/vuln/detail/CVE-2009-3767 |

## Vulnerability Record - CVE-2019-0190

| | |
|---|---|
| **CVE ID / NVD Reference** | **CVE-2019-0190**<br>https://nvd.nist.gov/vuln/detail/CVE-2019-0190 |
| **CVSS Score / Severity / Vector** | HIGH / CVSS 7.5 / v3.1<br>Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:N/ S:U/ C:N/ I:N/ A:H |
| **CWE ID(s)** | NVD-CWE-noinfo: No CWE information |
| **Affected Component + Version** | Cryptographic Library 3.0.13 |
| **Vulnerability Description** | NVD describes a denial-of-service condition that may let an attacker disrupt availability of the affected software path. |
| **Exploitability / Exposure Context** | Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features. |
| **Recommended Controls** | patch affected code paths, constrain exposed interfaces, and confirm watchdog/recovery behavior for availability loss. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review. |
| **VEX-Style Status** | Under Investigation |
| **Status** | Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM. |
| **Source / Reference** | extracted; NVD CVE API 2.0<br>https://nvd.nist.gov/vuln/detail/CVE-2019-0190 |

# 13. Electronic Records - 21 CFR Part 11

| Requirement | DocPlus+ Implementation | Part 11 |
|---|---|---|
| System Validation | DocPlus+ generation path is deterministic and testable through FastAPI route and PDF render checks. | 11.10(a) |
| Audit Trail | Generation record links CPL-20260708-2FCB, CAPA-20260708-42C9, timestamp, and agent trace events. | 11.10(e) |
| Access Controls | AI output is marked draft and requires QE/RA approval before controlled use. | 11.10(d) |
| Electronic Signatures | Signature block captures reviewer, approver, date, and meaning of signature. | 11.50 |
| Record Integrity | PDF is generated as a stable artifact with a deterministic document number and retrievable filename. | 11.10(c) |
| Human-Readable Copy | Selectable-text PDF is available through the document download endpoint. | 11.10(b) |
| Retention | Artifact is retained under output/pdf and can be indexed by M5 or a QMS repository. | 11.10(c) |

## Generation Log Summary

| | |
|---|---|
| **Generation Hash** | 2FCB42C9 |
| **Input Fingerprint** | 2FCB42C9AEAD3FD6 |
| **AI Draft Status** | Pending human quality and regulatory review |
| **Low Confidence Handling** | Evidence below 70% must be reviewed before release; current generated evidence remains visible with confidence values. |

# 14. AuditShadow and Trace Decay

## AuditShadow findings

| Finding(s) | Requirement(s) | Risk | Observation | Remediation |
|---|---|---|---|---|
| FINDING-014 | REQ-MEAS-032 | Critical | Linked test evidence was executed on v3.4.2, but the device current firmware is v3.4. | Repeat verification on the current firmware or justify equivalence with approved rationale. |

## Trace Decay alerts

| Requirement(s) | Test(s) | Tested FW | Required FW | Reason |
|---|---|---|---|---|
| REQ-MEAS-032 | TC-MEAS-032 | v3.4.2 | v3.4 | Test was executed on v3.4.2, while complaint firmware is v3.4. |

## 14.1 Audit Data Interpretation and Corrective Action

AuditShadow and Trace Decay use the live M1 traceability graph as the controlled source for verification status. When the report flags missing evidence, it means the requirement does not have a formal VERIFIED_BY test link in that graph. It does not mean M4 found no supporting material. In this run, M4 retrieved 12 candidate evidence item(s), but candidate snippets remain draft support until a Quality reviewer links them to a requirement, test record, firmware version, and controlled source identifier.

| Signal | What It Means | Source | Corrective Action |
|---|---|---|---|
| AuditShadow graph evidence gap | 0 requirement finding(s) lack a VERIFIED_BY test link in M1. | M1 readiness score and traceability matrix | Create or link verification test evidence to each requirement and rescore readiness. |
| M4 retrieved evidence | 12 candidate item(s) exist, but are not automatically controlled objective evidence. | M2 complaint pipeline and M4 evidence retrieval | Promote accepted snippets into controlled evidence nodes with requirement IDs, source, confidence, and firmware. |
| Trace Decay | 1 firmware freshness alert(s) require review before release or approval. | M2 Trace Decay over M1 firmware/test links | Rerun verification on current firmware or document an approved equivalence rationale. |
| CAPA linkage | Open CAPA or complaint risk can keep readiness from being clean even when evidence exists. | M1 graph relationships and M2 risk assessment | Close, justify, or explicitly reference CAPA impact before final regulatory approval. |

## Reason and Corrective Action

The root issue is a traceability-linkage gap rather than a simple absence of all evidence. Corrective action is to synchronize M4 retrieved evidence back into the M1 graph as controlled verification evidence, then rerun AuditShadow and Trace Decay. Until that link exists, the report correctly treats the material as candidate support and keeps the audit finding open.

# 14.2 AuditShadow Detail Appendix

| Finding(s) | Requirement(s) | Reference | Risk | Observation | Remediation |
|---|---|---|---|---|---|
| FINDING-014 | REQ-MEAS-032 | ISO 14971 | Critical | Linked test evidence was executed on v3.4.2, but the device current firmware is v3.4. | Repeat verification on the current firmware or justify equivalence with approved rationale. |

### Audit Interpretation

Major and critical findings should be triaged before report approval. A graph-linked evidence gap means the requirement is present in the M1 graph while still lacking a controlled verification test relationship acceptable for submission or inspection. Candidate RAG evidence must be reviewed, accepted, and linked before the finding can close.

## 14.3 Trace Decay Detail Appendix

| Requirement(s) | Test(s) | Tested Firmware | Required Firmware | Reason |
|---|---|---|---|---|
| REQ-MEAS-032 | TC-MEAS-032 | v3.4.2 | v3.4 | Test was executed on v3.4.2, while complaint firmware is v3.4. |

### Traceability Impact Statement

Trace Decay identifies verification evidence that may no longer represent the active or proposed firmware state. These alerts are not final nonconformities by themselves, but they are mandatory review inputs for release, CAPA, and audit-readiness decisions.

# 15. Cross-Standard Rendering Matrix

| Element | FDA / QMSR | EU MDR | ISO 13485 |
|---|---|---|---|
| Safety Requirement | Design input under QMSR; linked to verification evidence. | GSPR and Annex I safety/performance claim. | ISO 13485 Clause 7.3.3 design input. |
| Test Evidence | DHF verification record; current configuration/version required. | Annex II Section 6.1 V&V evidence. | Clause 7.3.6 design verification. |
| Complaint Record | Complaint file CPL-20260708-2FCB with malfunction assessment. | Post-market vigilance signal under Articles 83/87. | Complaint handling under Clause 8.2.2. |
| CAPA Record | CAPA-20260708-42C9 opened for Measurement and Sensor Performance. | Trend/CAPA input for PMS and vigilance review. | Corrective action under Clause 8.5.2. |
| Risk Documentation | Risk analysis tied to design controls and complaint file. | Benefit-risk update in Annex II Section 5. | Risk management continuity through production/post-production. |
| Product Documentation | Lifecycle impact for Measurement and Sensor Performance. | Design information and change-control evidence. | Validation and change-control evidence required. |

## Regulatory Lens Note

The same live graph and agent state are rendered under three regulatory lenses. This report therefore does not duplicate content blindly; it reformats the same evidence chain for FDA/QMSR, EU MDR, and ISO 13485 review contexts.

# 16. Document Control and Approval

| | |
|---|---|
| **Document No.** | DOCPLUS-M4-20260708-2FCB42C9 |
| **Revision** | A |
| **Revision Date** | 2026-07-08 |
| **Complaint ID** | CPL-20260708-2FCB |
| **CAPA ID** | CAPA-20260708-42C9 |
| **Prepared by** | DocPlus+ M4 Evidence and Documentation Layer |
| **Generated from** | Live M2 complaint pipeline, M1 graph context, M4 evidence retrieval, and device-level SBOM/NVD scan when attached |
| **Generated at** | 2026-07-08T09:48:14Z |
| **Agent status** | completed |
| **Classification** | Confidential - AI Generated Draft |
| **Next Review** | 2027-01-04 |

| Role | Name | Signature | Date |
|---|---|---|---|
| Quality Reviewer | | | |
| Regulatory Reviewer | | | |